Mean Fragility and Apparent Structure: A Cautionary Study Using Pascal, Delannoy, Motzkin, Catalan, and Worpitzky Triangles

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Abstract

Graphs often suggest structure before mathematics explains it. This note examines a small empirical case: ratios of arithmetic, RMS, geometric, and harmonic means formed from combinatorial triangle rows or diagonals. Pascal Fibonacci diagonals appear to show a striking period-two oscillation with an increasing envelope. At first sight this resembles the visual language of phase oscillation, group velocity, and even Feigenbaum-style period doubling. However, comparison with Delannoy, Motzkin, Catalan, and Worpitzky triangles shows that the phenomenon is not a generic artifact of averaging. The Pascal case has a specific generator and a specific parity asymmetry. The lesson is methodological: apparent graphical structure should be stress-tested across several central means and across nearby combinatorial families before one infers deep dynamics.

The caution

A graph may show something real. It may also show something made by the measuring instrument. In this setting the measuring instrument is not a physical detector, but a choice of mean. Given positive data

$$x_0, x_1, \ldots, x_{L-1},$$

we compare four central means:

$$A = \frac{1}{L}\sum_i x_i, \qquad Q = \sqrt{\frac{1}{L}\sum_i x_i^2},$$

$$G = \left(\prod_i x_i\right)^{1/L}, \qquad H = \frac{L}{\sum_i x_i^{-1}}.$$

These obey

$$H \le G \le A \le Q.$$

But they do not see the same object. The arithmetic mean sees additive mass. The RMS mean emphasises peaks. The geometric mean sees multiplicative bulk. The harmonic mean is highly sensitive to small boundary terms.

So the first methodological rule is:

$$\boxed{\text{If a pattern appears under only one mean, suspect the mean before the object.}}$$

The second rule is stronger:

$$\boxed{\text{If a pattern survives several means, vary the underlying combinatorial family.}}$$

1 The empirical observation

For each family we form a row or diagonal X_n, compute a mean M_n, and inspect the consecutive ratio

$$R_n = \frac{M_{n+1}}{M_n}.$$

The interactive for this exploration is at

$https://leelemacjames.github.io/pascal\_diagonal\_means\_learner.html.$

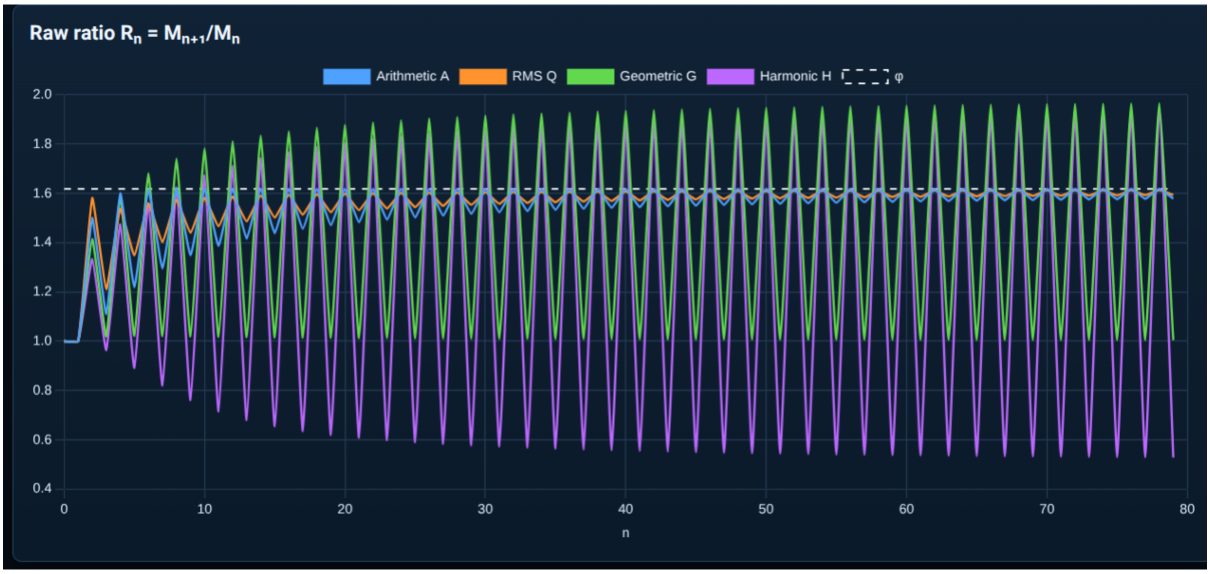


Figure 1: Graph of Pascal Fibonacci diagonals: ratios $R_n = M_{n+1}/M_n$ for arithmetic, RMS, geometric, and harmonic means suggesting an envelope, a phase oscillation, and a persistent two-cycle.

Three nearby controls

A graph can only be interpreted relative to alternatives so the Pascal Fibonacci diagonals are compared against three combinatorial families that retain important structural similarities:

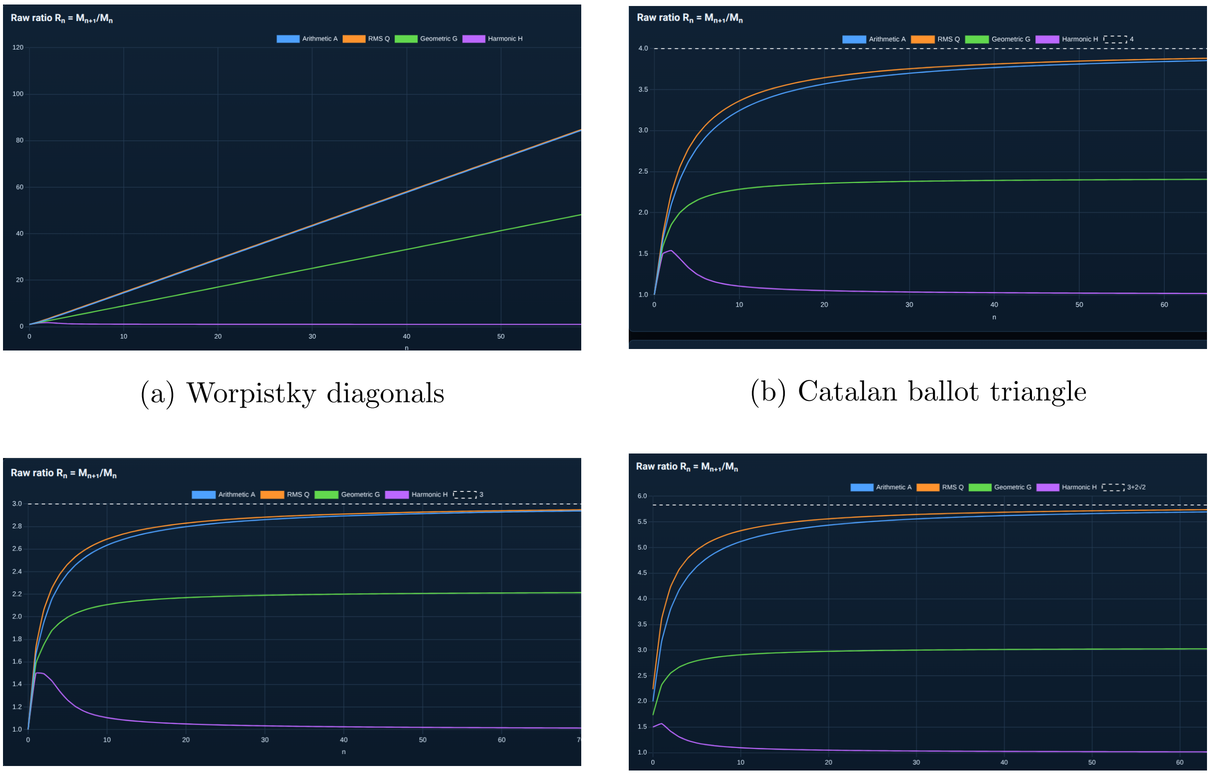
- Catalan ballot triangles, which preserve substantial binomial ancestry;
- Motzkin path triangles, which retain lattice-path interpretations;
- Delannoy triangles, which extend lattice paths by admitting diagonal steps.

Remarkably, all three exhibit qualitatively similar behaviour. Arithmetic and RMS ratios converge smoothly to their asymptotic growth constants. Geometric ratios converge to lower multiplicative-bulk limits. Harmonic ratios converge toward boundary-dominated limits close to unity. None exhibits the persistent parity oscillation observed in Pascal Fibonacci diagonals.

Catalan and Motzkin are closer controls because they retain path-counting or Pascal-adjacent structure. Delannoy is a broader lattice-path control. Worpitzky is a deliberately distant control involving Stirling and factorial scaling.

The Pascal graph is visually seductive and appears to contain:

- a fast initial envelope rise,



(a) Worpistky diagonals

(b) Catalan ballot triangle

(c) Motzkin path triangle

(d) Delannoy triangle

Figure 2: Comparison of ratio dynamics for arithmetic, RMS, geometric and harmonic means. Worpitsky, Catalan, Motzkin and Delannoy exhibit smooth convergence.

- a persistent phase oscillation,
- a limiting band,
- a period-two alternation,
- and a faint analogy with bifurcation diagrams or Feigenbaum period doubling.

The last analogy must be treated with caution. Feigenbaum period doubling concerns a family of dynamical systems with a control parameter. Here there is no obvious bifurcation parameter. The visual similarity is not yet mathematical evidence. A safer description is:

$$\boxed{\text{The Pascal ratios exhibit a stable parity split.}}$$

A small numerical window

The visual impression is already visible in a short numerical window. For Pascal Fibonacci diagonals, let

$$R_n^{(M)} = \frac{M_{n+1}}{M_n}.$$

n	$R_n^{(A)}$	$R_n^{(Q)}$	$R_n^{(G)}$	$R_n^{(H)}$
20	1.618034	1.598788	1.877862	1.804017
21	1.483198	1.531560	1.012594	0.607943
22	1.618034	1.600460	1.887749	1.818565
23	1.493570	1.538375	1.011829	0.598166

The arithmetic and RMS ratios remain close to the golden-ratio band. The geometric and harmonic ratios exhibit a much stronger even–odd alternation. This table is not a proof. It is a useful warning: the graph is showing something systematic, but different means are detecting different aspects of the data.

Robustness test across mean types

For Pascal Fibonacci diagonals, the arithmetic mean is especially transparent. Since

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1},$$

we have

$$A_n = \frac{F_{n+1}}{\lfloor n/2 \rfloor + 1}.$$

Therefore

$$\frac{A_{n+1}}{A_n} = \frac{F_{n+2}}{F_{n+1}} \cdot \frac{\lfloor n/2 \rfloor + 1}{\lfloor (n+1)/2 \rfloor + 1}.$$

The first factor tends to the golden ratio

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

The second factor carries the even–odd correction. Thus the arithmetic mean already predicts the main shape:

$$\boxed{R_n^{(A)} \approx \varphi \times \text{parity correction.}}$$

The RMS mean behaves similarly because it is dominated by the largest interior entries. The geometric mean exhibits a stronger two-cycle because it sees the whole multiplicative profile. The harmonic mean is the most violently parity-sensitive because it is pulled by small boundary terms.

The exact geometry of the Pascal parity split

The relevant Pascal diagonals are

$$D_n = \left\{ \binom{n-k}{k} : 0 \le k \le \lfloor n/2 \rfloor \right\}.$$

Writing them separately by parity gives

$$D_{2m} = \left(\binom{2m}{0}, \binom{2m-1}{1}, \binom{2m-2}{2}, \dots, \binom{m}{m} \right),$$

so

$$D_{2m} = \left(1,\, 2m-1,\, \binom{2m-2}{2},\, \dots,\, 1\right).$$

By contrast,

$$D_{2m+1} = \left(\binom{2m+1}{0}, \binom{2m}{1}, \binom{2m-1}{2}, \dots, \binom{m+1}{m} \right),$$

so

$$D_{2m+1} = \left(1,\, 2m,\, \binom{2m-1}{2},\, \dots,\, m+1\right).$$

Thus the earlier shorthand $D_{2m} = (1, \dots, 1)$ should not be read as saying that all entries are 1. The precise claim is that the two boundary entries of the even diagonal are both 1, whereas the right boundary of the odd diagonal is $m + 1$.

For the harmonic mean,

$$H_n = \frac{L_n}{\sum_i x_i^{-1}}.$$

The even case has two dominant reciprocal boundary contributions:

$$1^{-1} + 1^{-1} = 2.$$

The odd case has

$$1^{-1} + (m+1)^{-1} = 1 + \frac{1}{m+1} \to 1.$$

This explains why the harmonic mean has a persistent even–odd pulse. The oscillation is not a mysterious dynamical bifurcation. It is a boundary effect made visible by a reciprocal average.

The generator explanation

Define

$$D_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} x^k.$$

The ordinary generating function is

$$\sum_{n \geq 0} D_n(x) t^n = \frac{1}{1 - t - xt^2}.$$

At $x = 1$, this gives

$$D_n(1) = F_{n+1}.$$

Thus the additive mass of the diagonal is Fibonacci-governed. The denominator $1 - t - xt^2$ also explains why a second-order recurrence is present:

$$D_n(x) = D_{n-1}(x) + xD_{n-2}(x).$$

The graph therefore has an explanation:

$$\text{Fibonacci growth} + \text{second-order parity structure} + \text{mean sensitivity} = \text{observed ratio behaviour}.$$

Control families

1.1 Catalan and ballot rows

A closer Pascal-adjacent control is the Catalan or ballot triangle:

$$C(n,k) = \binom{n+k}{k} - \binom{n+k}{k-1}, \qquad 0 \leq k \leq n.$$

This family inherits binomial structure but imposes a non-crossing or ballot-type constraint. It is therefore a better control than Worpitzky if the question is whether Pascal ancestry alone produces the observed two-cycle.

1.2 Motzkin rows

Motzkin path rows may be generated by

$$T(n,k) = T(n-1,k-1) + T(n-1,k) + T(n-1,k+1),$$

with

$$T(0,0) = 1, \qquad T(n,k) = 0 \quad \text{for } k < 0.$$

These rows come from paths with up, level, and down steps. They test whether a path-family with a natural height boundary reproduces Pascal's persistent parity split.

1.3 Delannoy rows

Delannoy rows obey

$$D(n,k) = D(n-1,k) + D(n,k-1) + D(n-1,k-1),$$

with

$$D(n,0) = D(0,k) = 1.$$

The central Delannoy numbers have growth constant

$$3 + 2\sqrt{2}.$$

Empirically, arithmetic and RMS ratios approach this central-growth regime. The geometric mean approaches a lower multiplicative-bulk limit, and the harmonic mean approaches a boundary regime near 1. The important point is the absence of a persistent Pascal-like two-cycle.

1.4 Worpitzky rows

For Worpitzky rows,

$$W(n,k) = k!\, S(n+1,k+1).$$

These rows contain factorial and Stirling growth. Raw ratios are not expected to settle to a small constant in the same way. Instead, arithmetic and RMS ratios grow roughly monotonically. This is a useful distant control, but not the fairest nearby comparison.

The Worpitzky experiment answers a limited question:

Is the Pascal oscillation merely caused by taking means?

The answer appears to be no. But Catalan, Motzkin, and Delannoy are needed to test the stronger claim that the phenomenon is specific to Pascal Fibonacci diagonals.

2 Interpretation

The experiment separates three things that the first graph visually merged:

Envelope	:	early growth as the row or diagonal fills out,
phase oscillation	:	even–odd alternation,
asymptotic regime	:	late limiting behaviour of R_n.

The Pascal case has all three. Delannoy has clean limiting behaviour without a persistent parity band. Worpitzky has strong growth but no visible period-two structure. Motzkin and Catalan provide the fairer intermediate tests.

Therefore the Pascal phenomenon is real, but not mysterious in the dynamical-systems sense. It is a consequence of the Fibonacci diagonal generator and its parity-sensitive boundary structure.

A restrained conclusion

The original graph was neither meaningless or enough to build a foundational thesis.The responsible chain of inference is:

$$\text{graphical pattern} \longrightarrow \text{mean robustness} \longrightarrow \text{family robustness} \longrightarrow \text{generator explanation.}$$

For Pascal Fibonacci diagonals, the generator

$$\frac{1}{1-t-xt^2}$$

explains why Fibonacci growth appears, while the split between $2m$ and $2m+1$ explains why the ratios preserve an even–odd signature. The warning is therefore while graphs may deceive they are inclined to speak before they explain.

$$\boxed{\text{First see the structure, then try to destroy it.}}$$

If it survives, it may deserve a theorem.